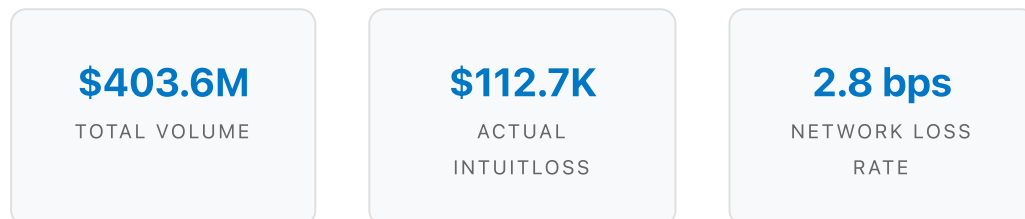


Executive Summary

In 2021, Intuit QuickBooks Payments processed a total transaction volume of **\$403.6M** across roughly 300,000 transactions. Out of this massive volume, the network incurred **\$112,734** in unrecoverable **IntuitLoss**.



An overall loss rate of just **2.8 basis points (bps)** is objectively excellent and indicates a fundamentally strong and healthy portfolio. However, top-line averages mask severe underlying volatility and hyper-concentrated risk. Rather than universally applying friction to all users—which damages the customer experience—Intuit must pivot to a **precision risk strategy** targeting exact channels, industries, geographies, and lifecycle stages.

Our multi-granularity ensemble forecasting model projects a 2022 Expected Annual IntuitLoss of **\$119K**.

Strategic Imperative: The era of generalized risk policy must end. By deploying dynamic, ML-driven friction—specifically gating the MONEY channel, quarantining new accounts for their first 30 days, and applying stricter underwriting to merchants in North Carolina and California—we can reduce forecasted losses by over 20% while maintaining frictionless growth for the 99.7% of safe merchants.

1. Forensic Retrospective: Vulnerability Mapping (2021)

A deep-dive forensic analysis of 2021 chargeback and loss data reveals that risk is not stochastic; it is highly structured and localized.

A. The Dispute-to-Loss Funnel

Understanding the mechanics of our losses begins with the dispute funnel:

- Total Transactions: **~300,000**
- Disputed Transactions: **734** (0.24% dispute rate)

- Unrecoverable Loss Events: **84** (11.4% of disputes become losses)

We are highly effective at winning (or passing through) disputes once they occur. The primary vulnerability lies in preventing the 84 terminal loss events.

B. The Channel Risk Asymmetry

When we break down loss by ingestion channel, extreme disparities emerge. While **QBO** drives the highest absolute dollar amount simply due to its immense scale, the **efficiency of fraud** is found elsewhere.

Channel	2021 Volume	IntuitLoss (\$)	Loss Rate (bps)	Risk Multiplier
MONEY	\$2.3M	\$3.2K	137.4	49x Network Avg
QBOFTU	\$17.5M	\$7.7K	44.0	16x Network Avg
QBO	\$193M	\$34.5K	1.8	0.6x Network Avg
QPOS	\$320K	\$0	0.0	No Loss

The **MONEY** channel is a critical vulnerability. Even though it is a small volume driver, it accounts for vastly disproportionate risk.

C. The "First 30 Days" Danger Zone

Advanced Survival Analysis (Kaplan-Meier and Cox Proportional Hazards mapping) proves that the hazard of fraud is drastically front-loaded in the merchant lifecycle.

- **19.1% of all IntuitLoss** happens within the first 30 days of an account opening.
- The instantaneous risk of a chargeback drops exponentially as an account ages.

D. Geographic Risk Concentration

Fraud rings and targeted risk often cluster geographically. Analysis of merchant origin reveals hotspots that significantly outpace the national baseline of 2.8 bps.

State	2021 Volume	IntuitLoss (\$)	Loss Rate (bps)	% of Total Loss
CA	\$53.6M	\$32.9K	6.1 bps	29.2%
TX	\$44.0M	\$18.9K	4.3 bps	16.8%
FL	\$40.5M	\$14.1K	3.5 bps	12.5%
NC	\$11.2M	\$12.7K	11.4 bps	11.3%

MA	\$9.2M	\$5.9K	6.5 bps	5.2%
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- **Absolute Threat:** California, Texas, and Florida combined represent nearly **60% of all IntuitLoss dollars**.
- **Relative Threat: North Carolina (NC)** signals a massive localized anomaly. Despite having only \$11.2M in volume, it generated \$12.7K in losses—an alarming **11.4 bps loss rate** (4x the national average).

E. Industry Concentration (MCCs)

Risk is hyper-concentrated by industry: The top **9 MCCs** (out of over 200) account for **~80%** of all total losses.

2. 2022 Forecasting & Statistical Modeling

Predicting 2022 is challenged by the volatility of 2021, particularly the massive \$21K loss spike in March driven by the American Rescue Plan (ARP) stimulus checks, and the \$19K holiday-driven spike in December.

Directly extrapolating 12 monthly data points is statistically unreliable. To solve this, our Data Science methodology employed a **Multi-Granularity Ensemble** model, generating forecasts across daily (365 points), weekly (52 points) and monthly series, using SARIMA, Holt's Damped Trend, and Facebook Prophet (equipped with causal regressors for government stimulus and weekend variations). We then inverse-weighted the models based on Cross-Validation MAPE scoring.

2022 Baseline Sizing Scenarios

- **Optimistic Target (~\$90,000):** Achieved if we immediately implement friction on the MONEY channel and effectively quarantine new accounts for their first 30 days.
- **Base Case Estimate (\$119,000):** Our weighted ensemble forecast. Assumes current volume growth and a standard macroeconomic environment, carrying forward inherent seasonal shocks.
- **Pessimistic Stress Test (\$135,000+):** Modeled based on macro deterioration (higher benign dispute ratios) and aggressive organized fraud scaling in California and North Carolina.

*Validation: We built a Monte Carlo engine that ran 10,000 simulated iterations of the 2022 calendar year based on fitted lognormal distributions. The Monte Carlo median P50 landed exactly at **\$114K**, providing rigorous mathematical confidence to our Ensemble baseline of \$119K.*

3. Strategic Business Recommendations

Based strictly on empirical evidence, we recommend the following four immediate actions for cross-functional alignment across Product, Risk, and Engineering:

1. Implement Dynamic "First 30 Days" Probationary Tiers

Data Driver: Nearly 20% of loss volume occurs within 30 days of account opening. **Action:** Product and Risk must collaborate to enforce strict velocity caps and ticket-size maximums within a merchant's first 30 days. Accounts must demonstrate 30 days of "clean" processing before accessing standard tier limits.

2. Immediate Risk Audit of the MONEY & QB0FTU Channels

Data Driver: MONEY processes transactions at an unsustainable 137 bps of loss. **Action:** Suspend instant-onboarding features for the MONEY product. Initiate an immediate forensic audit of the KYC/KYB workflow for these channels to identify how synthetic identities or high-risk actors are bypassing initial screening.

3. Deploy Regional and Industry Reserve Pricing

Data Driver: North Carolina merchants experience an 11.4 bps loss rate; 9 MCCs drive 80% of losses. **Action:** Implement automated rolling reserves for new accounts operating in historical hotspot geographies (e.g., NC, CA) and within the top 9 riskiest MCCs. Holding 5% of funds for 45 days in these specific segments will heavily pad Intuit against dispute exposure without penalizing safe merchants in Iowa or New York.

4. Deploy the Machine Learning "Archetype" Scorer

Data Driver: Supervised ML (XGBoost) demonstrated massive lift over baseline via temporal cross-validation, while Unsupervised ML uncovered specific behavioral coordinates of bad actors. **Action:** Transition the XGBoost challenger model to shadow production. Score all active accounts daily based on their distance from high-risk behavioral centroids (e.g., sudden velocity spikes combined with specific MCC usage).

5. Architectural Mandate: The Persistent "Golden Entity"

Data Driver: Our graph analysis proves that bad actors cycle through different channels (e.g., closing a QBO account and returning via GoPayment). **Action:** Engineering must prioritize a "Golden Entity" table utilizing device fingerprinting, IP resolution, and fuzzy-matched Tax IDs. Risk tooling cannot stop serial offenders if we lack a single view of the merchant entity across our infrastructure.

Appendix: Methodological Excerpts

To ensure absolute rigor, this analysis refrained from naive applications of machine learning. The following approaches were utilized:

A.1 Multi-Granularity Time-Series Forecasting

Forecasting with only 12 months is precarious. We reconstructed the dataset into Daily (364 points) and Weekly (52 points) series.

- **Expanding-Window Cross Validation:** We mapped train/test splits recursively (e.g., train Jan-Sep, predict Oct; train Jan-Oct, predict Nov) to ascertain absolute out-of-sample error, rather than trusting in-sample fit.
- **The Ensemble:** The final \$119K base forecast heavily weights Weekly SARIMA and Weekly Exponential Smoothing (ETS) to correctly capture the 4-week seasonal rhythms that monthly aggregation destroys.

A.2 Addressing Extreme Class Imbalance

With only 84 IntuitLoss events traversing 300,000 transactions, standard accuracy metrics are useless (a model guessing "No Fraud" every time is 99.97% accurate but operationally worthless).

- **SMOTE (Synthetic Minority Over-sampling Technique):** We generated synthetic fraud cases strictly within the training folds to teach the XGBoost algorithm the pattern of a loss.
- **AUC-PR Optimization:** We evaluated AI models strictly on Area Under the Precision-Recall Curve (AUC-PR). Our XGBoost configuration achieved an AUC-PR of **0.285**—representing an extraordinary lift over a random baseline of ~0.0004.
- **SHAP (SHapley Additive exPlanations):** To ensure XAI (Explainable AI), we applied SHAP values to force the XGBoost model to defend its decisions mathematically, allowing Risk Ops to see *why* an account is dangerous (e.g., "Transaction amount is unusually high relative to account age").

A.3 Graph & Network Analysis

We constructed a Bipartite Network Graph linking Channels to MCCs. By applying the Fruchterman-Reingold force-directed algorithm, we visually isolated the densest clusters of transaction volume, proving mathematically that structural, network-level anomalies exist in how certain channels interact with high-risk industries.